

Task 2: Detailed Plan Switch Analysis

CS24BTECH11037 CS24BTECH11036

Query 1

Plan Switch 1

Dates: 1992-01-07 -> 1992-01-08

Plan Before Switch:

```
[Level 0 Node]
- Operation: Aggregate
- Strategy: Plain
- Partial Mode: Simple
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Bitmap Heap Scan
  - Relation Name: lineitem
  - Parent Relationship: Outer
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Bitmap Index Scan
    - Index Name: idx_lineitem_shipdate
    - Parent Relationship: Outer
    - Parallel Aware: False
```

Plan After Switch:

```
[Level 0 Node]
- Operation: Aggregate
- Strategy: Plain
- Partial Mode: Simple
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Index Scan
  - Relation Name: lineitem
  - Index Name: idx_lineitem_shipdate
  - Parent Relationship: Outer
  - Parallel Aware: False
```

Plan Switch 2

Dates: 1992-01-27 -> 1992-01-28

Plan Before Switch:

```
[Level 0 Node]
- Operation: Aggregate
```

- Strategy: Plain
- Partial Mode: Simple
- Parallel Aware: False
- [Level 1 Node]
 - Operation: Index Scan
 - Relation Name: lineitem
 - Index Name: idx_lineitem_shipdate
 - Parent Relationship: Outer
 - Parallel Aware: False

Plan After Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Plain
 - Partial Mode: Finalize
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Gather
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 2 Node]
 - Operation: Aggregate
 - Parent Relationship: Outer
 - Strategy: Plain
 - Partial Mode: Partial
 - Parallel Aware: False
 - [Level 3 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_lineitem_shipdate
 - Parent Relationship: Outer
 - Parallel Aware: False

Plan Switch 3

Dates: 1992-03-28 -; 1992-03-29

Plan Before Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Plain
 - Partial Mode: Finalize
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Gather
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 2 Node]
 - Operation: Aggregate

- Parent Relationship: Outer
- Strategy: Plain
- Partial Mode: Partial
- Parallel Aware: False
- [Level 3 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_lineitem_shipdate
 - Parent Relationship: Outer
 - Parallel Aware: False

Plan After Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Plain
 - Partial Mode: Finalize
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Gather
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 2 Node]
 - Operation: Aggregate
 - Parent Relationship: Outer
 - Strategy: Plain
 - Partial Mode: Partial
 - Parallel Aware: False
- [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True

Query 2

Plan Switch 1

Dates: 1992-01-01 -> 1992-01-02

Plan Before Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Sorted
 - Partial Mode: Simple
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Sort
 - Parent Relationship: Outer

- Parallel Aware: False
- [Level 2 Node]
 - Operation: Nested Loop
 - Parent Relationship: Outer
 - Join Type: Semi
 - Parallel Aware: False
 - [Level 3 Node]
 - Operation: Index Scan
 - Relation Name: orders
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 3 Node]
 - Operation: Index Scan
 - Relation Name: lineitem
 - Index Name: lineitem_pkey
 - Parent Relationship: Inner
 - Parallel Aware: False

Plan After Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Sorted
 - Partial Mode: Finalize
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Gather Merge
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 2 Node]
 - Operation: Aggregate
 - Parent Relationship: Outer
 - Strategy: Sorted
 - Partial Mode: Partial
 - Parallel Aware: False
 - [Level 3 Node]
 - Operation: Sort
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 4 Node]
 - Operation: Nested Loop
 - Parent Relationship: Outer
 - Join Type: Semi
 - Parallel Aware: False
 - [Level 5 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 6 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False

```

[Level 5 Node]
- Operation: Index Scan
- Relation Name: lineitem
- Index Name: lineitem_pkey
- Parent Relationship: Inner
- Parallel Aware: False

```

Plan Switch 2

Dates: 1992-02-21 -> 1992-02-22

Plan Before Switch:

```

[Level 0 Node]
- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Gather Merge
  - Parent Relationship: Outer
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Aggregate
    - Parent Relationship: Outer
    - Strategy: Sorted
    - Partial Mode: Partial
    - Parallel Aware: False
      [Level 3 Node]
      - Operation: Sort
      - Parent Relationship: Outer
      - Parallel Aware: False
        [Level 4 Node]
        - Operation: Nested Loop
        - Parent Relationship: Outer
        - Join Type: Semi
        - Parallel Aware: False
          [Level 5 Node]
          - Operation: Bitmap Heap Scan
          - Relation Name: orders
          - Parent Relationship: Outer
          - Parallel Aware: True
            [Level 6 Node]
            - Operation: Bitmap Index Scan
            - Index Name: idx_orders_orderdate
            - Parent Relationship: Outer
            - Parallel Aware: False
          [Level 5 Node]
          - Operation: Index Scan
          - Relation Name: lineitem
          - Index Name: lineitem_pkey
          - Parent Relationship: Inner
          - Parallel Aware: False

```

Plan After Switch:

```

[Level 0 Node]
- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Gather Merge
  - Parent Relationship: Outer
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Aggregate
    - Parent Relationship: Outer
    - Strategy: Sorted
    - Partial Mode: Partial
    - Parallel Aware: False
      [Level 3 Node]
      - Operation: Sort
      - Parent Relationship: Outer
      - Parallel Aware: False
        [Level 4 Node]
        - Operation: Nested Loop
        - Parent Relationship: Outer
        - Join Type: Semi
        - Parallel Aware: False
          [Level 5 Node]
          - Operation: Seq Scan
          - Relation Name: orders
          - Parent Relationship: Outer
          - Parallel Aware: True
            [Level 5 Node]
            - Operation: Index Scan
            - Relation Name: lineitem
            - Index Name: lineitem_pkey
            - Parent Relationship: Inner
            - Parallel Aware: False

```

Plan Switch 3

Dates: 1992-03-18 -; 1992-03-19

Plan Before Switch:

```

[Level 0 Node]
- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Gather Merge
  - Parent Relationship: Outer
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Aggregate
    - Parent Relationship: Outer

```

- Strategy: Sorted
- Partial Mode: Partial
- Parallel Aware: False
- [Level 3 Node]
 - Operation: Sort
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 4 Node]
 - Operation: Nested Loop
 - Parent Relationship: Outer
 - Join Type: Semi
 - Parallel Aware: False
 - [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 5 Node]
 - Operation: Index Scan
 - Relation Name: lineitem
 - Index Name: lineitem_pkey
 - Parent Relationship: Inner
 - Parallel Aware: False

Plan After Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Sorted
 - Partial Mode: Finalize
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Gather Merge
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 2 Node]
 - Operation: Aggregate
 - Parent Relationship: Outer
 - Strategy: Sorted
 - Partial Mode: Partial
 - Parallel Aware: False
 - [Level 3 Node]
 - Operation: Sort
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 4 Node]
 - Operation: Nested Loop
 - Parent Relationship: Outer
 - Join Type: Semi
 - Parallel Aware: False
 - [Level 5 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True

```

[Level 6 Node]
- Operation: Bitmap Index Scan
- Index Name: idx_orders_orderdate
- Parent Relationship: Outer
- Parallel Aware: False
[Level 5 Node]
- Operation: Index Scan
- Relation Name: lineitem
- Index Name: lineitem_pkey
- Parent Relationship: Inner
- Parallel Aware: False

```

Plan Switch 4

Dates: 1992-05-24 -; 1992-05-25

Plan Before Switch:

```

[Level 0 Node]
- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False
[Level 1 Node]
- Operation: Gather Merge
- Parent Relationship: Outer
- Parallel Aware: False
[Level 2 Node]
- Operation: Aggregate
- Parent Relationship: Outer
- Strategy: Sorted
- Partial Mode: Partial
- Parallel Aware: False
[Level 3 Node]
- Operation: Sort
- Parent Relationship: Outer
- Parallel Aware: False
[Level 4 Node]
- Operation: Nested Loop
- Parent Relationship: Outer
- Join Type: Semi
- Parallel Aware: False
[Level 5 Node]
- Operation: Bitmap Heap Scan
- Relation Name: orders
- Parent Relationship: Outer
- Parallel Aware: True
[Level 6 Node]
- Operation: Bitmap Index Scan
- Index Name: idx_orders_orderdate
- Parent Relationship: Outer
- Parallel Aware: False
[Level 5 Node]
- Operation: Index Scan

```

- Relation Name: lineitem
- Index Name: lineitem_pkey
- Parent Relationship: Inner
- Parallel Aware: False

Plan After Switch:

[Level 0 Node]

- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False

[Level 1 Node]

- Operation: Gather Merge
- Parent Relationship: Outer
- Parallel Aware: False

[Level 2 Node]

- Operation: Aggregate
- Parent Relationship: Outer
- Strategy: Sorted
- Partial Mode: Partial
- Parallel Aware: False

[Level 3 Node]

- Operation: Sort
- Parent Relationship: Outer
- Parallel Aware: False

[Level 4 Node]

- Operation: Nested Loop
- Parent Relationship: Outer
- Join Type: Semi
- Parallel Aware: False

[Level 5 Node]

- Operation: Seq Scan
- Relation Name: orders
- Parent Relationship: Outer
- Parallel Aware: True

[Level 5 Node]

- Operation: Index Scan
- Relation Name: lineitem
- Index Name: lineitem_pkey
- Parent Relationship: Inner
- Parallel Aware: False

Plan Switch 5

Dates: 1992-05-25 -; 1992-05-26

Plan Before Switch:

[Level 0 Node]

- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False

[Level 1 Node]

- Operation: Gather Merge
- Parent Relationship: Outer
- Parallel Aware: False
- [Level 2 Node]
 - Operation: Aggregate
 - Parent Relationship: Outer
 - Strategy: Sorted
 - Partial Mode: Partial
 - Parallel Aware: False
- [Level 3 Node]
 - Operation: Sort
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 4 Node]
 - Operation: Nested Loop
 - Parent Relationship: Outer
 - Join Type: Semi
 - Parallel Aware: False
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 5 Node]
 - Operation: Index Scan
 - Relation Name: lineitem
 - Index Name: lineitem_pkey
 - Parent Relationship: Inner
 - Parallel Aware: False

Plan After Switch:

- [Level 0 Node]
 - Operation: Aggregate
 - Strategy: Sorted
 - Partial Mode: Finalize
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Gather Merge
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 2 Node]
 - Operation: Aggregate
 - Parent Relationship: Outer
 - Strategy: Sorted
 - Partial Mode: Partial
 - Parallel Aware: False
- [Level 3 Node]
 - Operation: Sort
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 4 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Semi

- Parallel Aware: True
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 5 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 6 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan Switch 6

Dates: 1998-01-09 -; 1998-01-10

Plan Before Switch:

```
[Level 0 Node]
- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False
[Level 1 Node]
- Operation: Gather Merge
- Parent Relationship: Outer
- Parallel Aware: False
[Level 2 Node]
- Operation: Aggregate
- Parent Relationship: Outer
- Strategy: Sorted
- Partial Mode: Partial
- Parallel Aware: False
[Level 3 Node]
- Operation: Sort
- Parent Relationship: Outer
- Parallel Aware: False
[Level 4 Node]
- Operation: Hash Join
- Parent Relationship: Outer
- Join Type: Semi
- Parallel Aware: True
[Level 5 Node]
- Operation: Seq Scan
- Relation Name: orders
- Parent Relationship: Outer
- Parallel Aware: True
[Level 5 Node]
- Operation: Hash
- Parent Relationship: Inner
```

- Parallel Aware: True
- [Level 6 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan After Switch:

```
[Level 0 Node]
- Operation: Aggregate
- Strategy: Sorted
- Partial Mode: Finalize
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Gather Merge
  - Parent Relationship: Outer
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Sort
    - Parent Relationship: Outer
    - Parallel Aware: False
      [Level 3 Node]
      - Operation: Aggregate
      - Parent Relationship: Outer
      - Strategy: Hashed
      - Partial Mode: Partial
      - Parallel Aware: False
        [Level 4 Node]
        - Operation: Hash Join
        - Parent Relationship: Outer
        - Join Type: Semi
        - Parallel Aware: True
          [Level 5 Node]
          - Operation: Seq Scan
          - Relation Name: orders
          - Parent Relationship: Outer
          - Parallel Aware: True
            [Level 5 Node]
            - Operation: Hash
            - Parent Relationship: Inner
            - Parallel Aware: True
              [Level 6 Node]
              - Operation: Seq Scan
              - Relation Name: lineitem
              - Parent Relationship: Outer
              - Parallel Aware: True
```

Query 3

Plan Switch 1

Dates: 1992-01-04 -; 1992-01-05

Plan Before Switch:

```
[Level 0 Node]
- Operation: Gather
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Nested Loop
  - Parent Relationship: Outer
  - Join Type: Inner
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Nested Loop
    - Parent Relationship: Outer
    - Join Type: Inner
    - Parallel Aware: False
      [Level 3 Node]
      - Operation: Bitmap Heap Scan
      - Relation Name: orders
      - Parent Relationship: Outer
      - Parallel Aware: True
        [Level 4 Node]
        - Operation: Bitmap Index Scan
        - Index Name: idx_orders_orderdate
        - Parent Relationship: Outer
        - Parallel Aware: False
      [Level 3 Node]
      - Operation: Index Scan
      - Relation Name: customer
      - Index Name: customer_pkey
      - Parent Relationship: Inner
      - Parallel Aware: False
    [Level 2 Node]
    - Operation: Index Scan
    - Relation Name: lineitem
    - Index Name: lineitem_pkey
    - Parent Relationship: Inner
    - Parallel Aware: False
```

Plan After Switch:

```
[Level 0 Node]
- Operation: Gather
- Parallel Aware: False
  [Level 1 Node]
  - Operation: Nested Loop
  - Parent Relationship: Outer
  - Join Type: Inner
  - Parallel Aware: False
    [Level 2 Node]
    - Operation: Hash Join
    - Parent Relationship: Outer
    - Join Type: Inner
    - Parallel Aware: True
      [Level 3 Node]
      - Operation: Bitmap Heap Scan
```

- Relation Name: orders
- Parent Relationship: Outer
- Parallel Aware: True
- [Level 4 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 3 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Index Scan
 - Relation Name: lineitem
 - Index Name: lineitem_pkey
 - Parent Relationship: Inner
 - Parallel Aware: False

Plan Switch 2

Dates: 1992-01-19 -; 1992-01-20

Plan Before Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Nested Loop
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: False
 - [Level 2 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 3 Node]

- Operation: Hash
- Parent Relationship: Inner
- Parallel Aware: True
- [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Index Scan
 - Relation Name: lineitem
 - Index Name: lineitem_pkey
 - Parent Relationship: Inner
 - Parallel Aware: False

Plan After Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 5 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False
- [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Seq Scan

- Relation Name: customer
- Parent Relationship: Outer
- Parallel Aware: True

Plan Switch 3

Dates: 1992-02-18 -; 1992-02-19

Plan Before Switch:

[Level 0 Node]

- Operation: Gather
- Parallel Aware: False

[Level 1 Node]

- Operation: Hash Join
- Parent Relationship: Outer
- Join Type: Inner
- Parallel Aware: True

[Level 2 Node]

- Operation: Hash Join
- Parent Relationship: Outer
- Join Type: Inner
- Parallel Aware: True

[Level 3 Node]

- Operation: Seq Scan
- Relation Name: lineitem
- Parent Relationship: Outer
- Parallel Aware: True

[Level 3 Node]

- Operation: Hash
- Parent Relationship: Inner
- Parallel Aware: True

[Level 4 Node]

- Operation: Bitmap Heap Scan
- Relation Name: orders
- Parent Relationship: Outer
- Parallel Aware: True

[Level 5 Node]

- Operation: Bitmap Index Scan
- Index Name: idx_orders_orderdate
- Parent Relationship: Outer
- Parallel Aware: False

[Level 2 Node]

- Operation: Hash
- Parent Relationship: Inner
- Parallel Aware: True

[Level 3 Node]

- Operation: Seq Scan
- Relation Name: customer
- Parent Relationship: Outer
- Parallel Aware: True

Plan After Switch:

[Level 0 Node]

- Operation: Gather
- Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan Switch 4

Dates: 1992-03-15 -; 1992-03-16

Plan Before Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner

- Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan After Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 4 Node]

- Operation: Hash
- Parent Relationship: Inner
- Parallel Aware: True
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan Switch 5

Dates: 1992-03-20 -; 1992-03-21

Plan Before Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan After Switch:

- [Level 0 Node]

- Operation: Gather
- Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 5 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 4 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan Switch 6

Dates: 1992-05-26 -> 1992-05-27

Plan Before Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner

- Parallel Aware: True
 - [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Bitmap Heap Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 5 Node]
 - Operation: Bitmap Index Scan
 - Index Name: idx_orders_orderdate
 - Parent Relationship: Outer
 - Parallel Aware: False
 - [Level 4 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan After Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 3 Node]

- Operation: Hash Join
- Parent Relationship: Outer
- Join Type: Inner
- Parallel Aware: True
- [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan Switch 7

Dates: 1992-12-15 -; 1992-12-16

Plan Before Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
- [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 3 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Hash

- Parent Relationship: Inner
- Parallel Aware: True
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan After Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan Switch 8

Dates: 1993-03-13 -; 1993-03-14

Plan Before Switch:

- [Level 0 Node]
 - Operation: Gather

- Parallel Aware: False
 - [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 3 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True

Plan After Switch:

- [Level 0 Node]
 - Operation: Gather
 - Parallel Aware: False
 - [Level 1 Node]
 - Operation: Hash Join
 - Parent Relationship: Outer
 - Join Type: Inner
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Seq Scan
 - Relation Name: lineitem
 - Parent Relationship: Outer
 - Parallel Aware: True
 - [Level 2 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
 - [Level 3 Node]

- Operation: Hash Join
- Parent Relationship: Outer
- Join Type: Inner
- Parallel Aware: True
- [Level 4 Node]
 - Operation: Seq Scan
 - Relation Name: orders
 - Parent Relationship: Outer
 - Parallel Aware: True
- [Level 4 Node]
 - Operation: Hash
 - Parent Relationship: Inner
 - Parallel Aware: True
- [Level 5 Node]
 - Operation: Seq Scan
 - Relation Name: customer
 - Parent Relationship: Outer
 - Parallel Aware: True